FLEET TEAM DIGEST

737MAX-FTD-32-17016

Issue Title : Improving Service Experience With Wheel Tie Bolts

Airplane Model	ATA	Minor Model(s)
737MAX	3240-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG, 757, 737	9E991

Originated Date	Last Revised Date	Created On
05/11/2017	09/18/2026	07/30/2026

Estimated Completion Date	Next Update Date

Revision Description

Provide guidance when two or more tie bolts fail during service.

Applicability

All 737MAX airplanes.

Description

Operators occasionally report fractured main gear wheel tie bolts on in-service airplanes. In some cases, the bolts are found missing during walk-arounds, resulting in flight cancellations or dispatch delays.

In other cases, the fractured bolts are not immediately found, which can lead to multiple fractured bolts. Once a wheel has several consecutive fractured bolts, the tire pressure can escape, resulting in a tire failure. In some cases, multiple tie bolt fractures has resulted in wheel fracture.

This article discusses ways to improve service experience with wheel tie bolts.

Background

Tie bolt fracture has posed sporadic maintenance problems on many different airplane models. Due to the rolling of the wheel, the number of fatigue cycles on wheel tie bolts accumulates very quickly. A single airplane flight cycle translates into several thousand fatigue cycles on each wheel tie bolt. As a result, tie bolts are prone to fatigue fractures. As soon as one bolt fractures, it transfers its load onto adjacent bolts, which greatly increases their fatigue damage, leading to a snow-balling effect of broken bolts. Corrosion pits on steel bolts also greatly increases the likelihood of fractures.

It is our experience that tie bolts can fracture for numerous reasons:

FATIGUE INITIATING FROM CORROSION PITS

Many bolts we examine have fatigue fractures that are found to have initiated from corrosion pits. Typically, these bolts are found to have visible corrosion pits in many locations on the bolt shank and threads. See the attached photo for an example of a severely corroded bolt that experienced a fatigue fracture, initiating from a corrosion pit.

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PURE FATIGUE

Some of the broken bolts Boeing has examined have fractured as a result of pure fatigue. This is a more difficult problem to resolve since the bolts would likely have passed NDT checks at the previous installation.

WHEEL MATING FACE FLATNESS:

There is mounting evidence that the flatness and condition of the mating surfaces of the wheel halves highly influences tie bolt life. One theory is that the mating faces on the wheel halves wear in such a way that the faces become uneven or tapered. This puts the bolt in a bending mode when tightened, thereby dramatically reducing bolt fatigue life.

POOR MAINTENANCE TECHNIQUES

Poor maintenance techniques such as incorrect torquing procedure, incorrect washer orientation, inadequate lubrication, inadequate component inspection, etc. have led to bolt fractures.

BOLT MANUFACTURING DEFECTS

In a few cases, it has been found that a batch of bolts or nuts had a manufacturing problem that led to fractures. However, this appears to be very rare since metallurgical analysis of damaged bolts rarely finds problems with material or heat treat.

Status

Closed.

Interim Action

N/A

Final Action

N/A

Milestones

FTD closed.

Operator Action

The following suggestions may help improve tie bolt service life:

1) Boeing recommends tie bolt life limits. Some operators report that with a life limit, they simply perform detailed visual inspections of the bolts at each tire change rather than slow and marginally effective formal NDT (such as magnetic particle, ultrasonic, or dye penetrant inspections). The reduced NDT burden and improved service experience can partially or fully offset the cost of bolt replacement with a life limit. A life limit requires some sort of coding to be placed on the bolt and will require a fleet purge of old bolts (usually a date or code is vibro-engraved on the end of the bolt). Establishing the proper life limit may take some experience to achieve acceptable results. We are aware of life limits ranging from 2500 flight cycles to 10000 flight cycles.

2) The wheel suppliers indicate that snug-angle methods to tighten bolts result in a more consistent bolt pre-load, which increases bolt fatigue life. (Snug-angle is sometimes called torque-angle.) Operators may wish to implement snug-angle methods if approved for the particular wheel part number. Most automated wheel tie bolt torquing equipment (such as Atlas Copco

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machines) have snug-angle capability. Also, there are some new low cost hand-held torque wrenches available that have snug-angle capability. Snap-On, KD tools, Tohnichi, CDI and others manufacture these wrenches.

3) Visually examine the bolts at each tire change using magnification. Any visible signs of corrosion pits or galling on the threads is cause for rejection.

4) Ensure that the washers for the bolts are not galled or cupped. Galled or cupped washers prevents the proper preload to be reached which dramatically reduces the bolt fatigue life.

5) If the bolt washers contain a countersink, ensure that the countersink is oriented correctly during wheel assembly.

6) Ensure that the nuts do not have galling or excessive wear. Operators may wish to replace the nuts on a life limit schedule, such as after 10 uses.

7) Boeing suggests that operators ensure that traces of dried thread lube and other dirt is thoroughly cleaned from the threads of the bolts and nuts during tire changes. Plastic bristle brushes are recommended if mechanical cleaning of the threads is required. Do not use metallic wire wheels on plated bolts since this quickly remove the plating (e.g. cadmium), leading to corrosion.

8) Ensure that the proper thread lubrication is liberally applied to the pertinent surfaces of the bolts, nuts, and washers. Testing shows that an insufficient quantity of thread lubricant can substantially reduce bolt preload which affects bolt fatigue life.

9) Several wheel suppliers have performed tests that demonstrate that wheel mating face flatness and surface condition of the tie bolt bosses affects bolt fatigue life. If there is tapered or uneven wear on the wheel half mating faces, it puts the bolts into a bending mode, which affects bolt fatigue life. Not all wheels currently have inspection criteria or repair schemes for wheel half mating faces. Operators should contact their wheel suppliers if the CMM does not have criteria.

10) Airlines should train their maintenance personnel and flight crews to look for broken tie bolts during airplane turn-arounds. This is somewhat difficult on 737 outboard wheels due to the aerodynamic hubcap. However, expensive airplane damage from a tire failure can be avoided if a wheel with a single broken bolt is discovered early, before multiple bolts fracture.

During in-service operations, if two or more adjacent tie bolts are missing, broken, or can freely rotate by hand, the mating wheel halves likely experienced fatigue damage. When this condition is observed, remove all tire pressure and remove the wheel-tire assembly per applicable AMM instructions. The wheel shop must disposition the wheel assembly per Supplier CMM instructions when tie bolt failures occur.

If Supplier CMM instructions are not followed when there are multiple missing/broken/loose tie bolts, the wheel assemblies can be at risk of high-energy wheel fracture as described within the reference FTD Article.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Economic (FTE)	FTE-32-12026		
Fleet Team Digest Article (FTD)	737MAX-FTD-32-18002		

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board
- In Service Occurrences : null

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Attachments:

a) Typical Fatigue Fracture

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